

InDeep^{Net}: a web platform for predicting functional binding sites in proteins using InDeep

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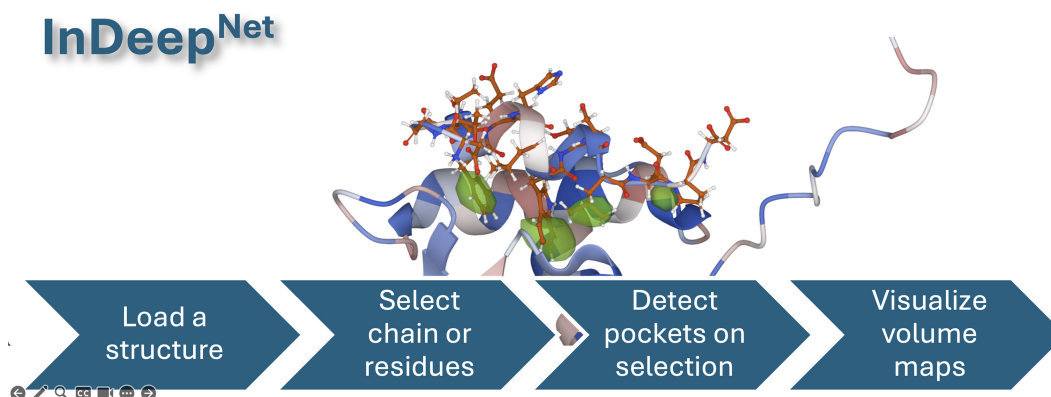
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Abstract

Predicting functional binding sites in proteins is crucial for understanding protein–protein interactions (PPIs) and identifying drug targets. While various computational approaches exist, many fail to assess PPI ligandability, which often involves conformational changes. We introduce InDeep^{Net}, a web-based platform integrating InDeep, a deep-learning model for binding site prediction, with InDeepHolo, which evaluates a site's propensity to adopt a ligand-bound (holo) conformation. InDeep^{Net} provides an intuitive interface for researchers to upload protein structures from in-house data, the Protein Data Bank (PDB), or AlphaFold, predicting potential binding sites for proteins or small molecules. Results are presented as interactive 3D visualizations via Mol*, facilitating structural analysis. With InDeep^{Holo}, the platform helps select conformations optimal for small-molecule binding, improving structure-based drug design. Accessible at <https://indeep-net.gpu.pasteur.cloud/>, InDeep^{Net} removes the need for specialized coding skills or high-performance computing, making advanced predictive models widely available. By streamlining PPI target assessment and ligandability prediction, it assists research and supports therapeutic development targeting PPIs.

Graphical abstract



Introduction

Protein–protein interactions (PPIs) play fundamental roles in a wide range of biological processes, mediating signal transduction, enzymatic regulation, and macromolecular assembly. Given their critical involvement in disease pathways, PPIs have become increasingly attractive targets for drug discovery, with numerous successful small-molecule inhibitors emerging in recent years [1, 2]. However, targeting PPIs remains challenging due to their inherently large and often shallow binding interfaces, which lack well-defined cavities typically seen in conventional drug targets [3] such as enzymes or G-protein-

coupled receptors. As a result, assessing the ligandability of PPIs and identifying suitable binding sites for small-molecule intervention require specialized computational tools.

Computational methods for PPI target assessment

Numerous computational approaches have been developed to assess the druggability of proteins and identify binding sites. Traditional geometric-based pocket detection methods such as Fpocket [4, 5], VolSite [6], and mkgridXf [7] have demonstrated effectiveness in detecting small-molecule binding sites,

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particularly for well-structured targets. Fragment-based approaches like FTMap [8] provide insights into potential hot spots by evaluating the interaction of small molecular fragments with protein surfaces. Meanwhile, deep learning-based tools such as DeepSite [9], Kalasanty [10], and DeepSurf [11] have further improved binding site detection by leveraging neural networks trained on protein structural data. While these tools provide valuable predictions, few have been specifically designed to assess the ligandability of PPIs. PPIs often undergo significant conformational rearrangements upon ligand binding, and static structure-based methods alone may fail to identify suitable pockets in their apo or complexed states [12, 13]. To address this, ensemble docking and molecular dynamics (MD) simulations have been used to sample alternative conformations and identify transiently ligandable states [14, 15]. However, selecting holo-like conformations from MD trajectories remains a challenge, as most scoring methods were developed for traditional drug targets and do not account for the unique properties of PPI interfaces [16, 17].

InDeep^{Net}: a web server for PPI binding site analysis

To improve PPI target assessment and facilitate access to predictive tools, we introduce InDeep^{Net}, an interactive web server integrating (Fig. 1) our tool InDeep [18], a deep-learning-based model for functional binding site prediction, and a newly developed tool, InDeep^{Holo}, designed to evaluate the holo-likeness of binding sites. InDeep was originally developed to predict ligandable and epitope-binding sites within PPIs, leveraging a 3D convolutional neural network (CNN) trained on curated datasets from inhibitor protein–protein interactions (iPPIs)-(DB) database [19, 20]. While InDeep successfully identifies binding pockets at or near PPI interfaces even within APO (unbound) conformations, the conformational flexibility of these sites remains a critical factor in their suitability for small-molecule binding.

To address this limitation, we introduce InDeep^{Holo}, a deep-learning model trained to assess whether a given protein conformation is likely to resemble a ligand-bound (holo) state. InDeep^{Holo} builds upon a dataset of known PPI inhibitors and their corresponding bound/unbound structures, leveraging machine learning to predict root mean square deviation (RMSD) from a holo conformation. While a comprehensive study on InDeep^{Holo} is forthcoming, we provide an initial validation of its performance within InDeep^{Net}, demonstrating its ability to prioritize holo-like conformations for molecular docking simulations.

By integrating these predictive tools into a web server, InDeep^{Net} enables researchers to:

1. Upload and analyze protein structures to predict binding sites relevant to drug discovery.
2. Evaluate the ligandability of these sites using deep-learning-based predictions.
3. Assess holo-likeness to prioritize conformations for molecular docking and virtual screening.

With InDeep^{Net}, we provide a publicly accessible platform that enhances PPI-targeted drug discovery by offering both binding site predictions and holo-likeness assessment in a single interface. As targeting PPIs remains a significant challenge in drug development, InDeep^{Net} represents an important step toward making advanced predictive models more accessible and actionable for researchers worldwide.

Materials and methods

InDeep protocol

InDeep is a deep-learning-based tool for predicting functional binding sites in proteins, focusing on iPPIs and epitope binding sites. It employs a 3D fully convolutional U-Net architecture, trained on structural data from iPPI-DB, using a voxel-based protein representation categorized into five functional atom types (α -carbon, hydrogen bond donors/acceptors, hydrophobic/aromatic, positively charged, negatively charged). Through multi-task learning, InDeep simultaneously predicts ligandable binding sites (PL Protein-Ligand interactions) for small-molecule binding and interactability patches (HD - HeteroDimer interactions) for PPIs. After processing through shared convolutional layers, it splits into two branches: one using sigmoid activation for ligandability prediction and the other using softmax activation for interactability prediction. Benchmarking on several datasets demonstrated that InDeep outperforms state of the art tools for ligand binding prediction and for interactability detection. It proved particularly effective in detecting iPPI-specific binding pockets, performing well on both holo (bound) and apo (unbound) structures, and in epitope binding site prediction and spatial localization of protein partners.

InDeep^{Holo} protocol

InDeep^{Holo} predicts holo-likeness by training on structural ensembles derived from the same dataset as InDeep, sourced from iPPI-DB [19] (target-centric mode), and split based on CATH folds to avoid structural overlap. The dataset includes hetero-dimeric complexes and compound-bound protein complexes, where ligands bind at PPI interfaces. Structural ensembles were generated using Rosetta's backrub protocol, running 1000 000 Monte Carlo steps at 2.5 kT, producing up to 16 models per protein. Binding site residues were identified based on changes in solvent-accessible surface upon ligand binding, measured using Naccess software. The binding sites were then encoded into 3D grids of size 20 Å³ with 1 Å voxel resolution, where atomic coordinates were categorized into five functional types (α -carbon, hydrogen bond donors/acceptors, hydrophobic/aromatic, positively charged, negatively charged) and transformed into probability distributions to create input tensors for deep learning. A CNN was used, consisting of three convolutional blocks (3D convolutions, BatchNorm, ELU activation) alternating with Max-Pooling layers, followed by fully connected layers. The model was trained for 500 epochs using mean squared error loss, Adam optimization, and 90-degree rotational data augmentation. Model validation relied on computing heavy-atom binding site RMSD against experimental holo structures, using α -carbon alignment for comparison, with performance assessed through Pearson's correlation coefficient and mean absolute error calculated via Scipy and Sklearn Python libraries (Supplementary Fig. S1).

Server Functionalities and Usage: The InDeep^{Net} server offers a range of functionalities to facilitate InDeep usability and InDeep results visualization and analysis. Users can upload structures and trajectories from public databases such as RCSB [21] (<http://www.rcsb.org/>), PDBe [22] (<https://www.ebi.ac.uk/pdbe/>), or AlphaFold [23] (<https://www.alphafold.ebi.ac.uk/>), or from their local computer. The server supports the upload of PDB structures and dynamic trajectories and provides a range of visualization tools powered by the Mol*

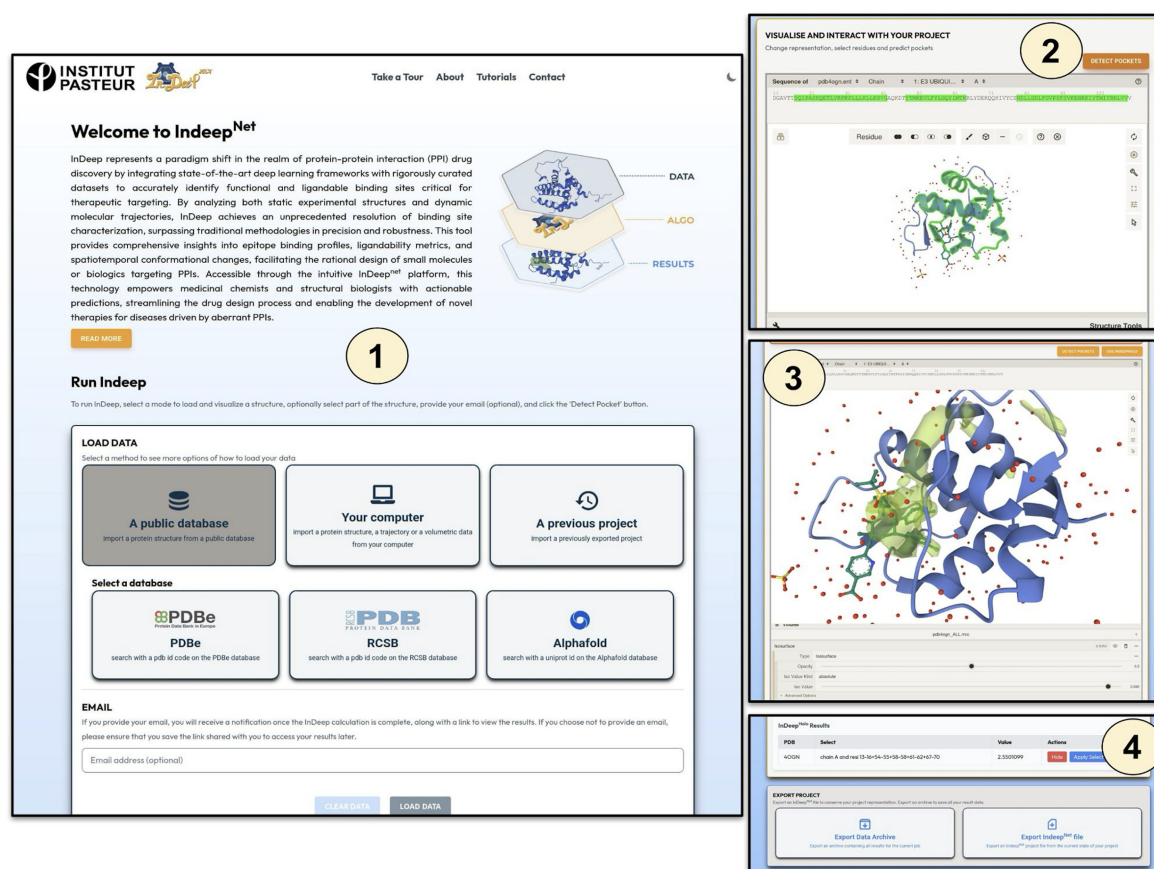


Figure 1. InDeep^{Net} website interface, illustrating the following steps: (i) Loading a structure from your computer or a public database; (ii) visualizing and interacting with the structure using Mol*, and selecting the residues for running an InDeep calculation; (iii) visualizing and interacting with InDeep results, and running an InDeep^{Holo} calculation; and (iv) visualizing InDeep^{Holo} results and exporting an archive or a Mol* visualization state.

plugin [24] (<https://molstar.org/>), including structure superposition, residue selection, and more.

InDeep Calculation: The InDeep^{Net} server handles InDeep jobs with all the necessary configuration for GPU calculation and parameters selection. Users can provide their email address (but it is not mandatory) to receive updates on the status of their job or save the given web link to retrieve their results later. When the job is complete, all results are loaded into the frontend application, and users can interact with the Mol* plugin to visualize and manipulate the results. Users can also load other structures and superimpose them with their results, by example some structures already liganded.

Results Visualization and Conservation: Once a job is complete, users can download an archive of results containing all pocket files calculated, pockets volume files and provided structure or trajectory. Additionally, users can save a project state file with the configuration of the visualization, allowing them to reload the visualization configuration for approximately 30 days and retrieve the same visualization state saved. To optimize calculation time and conserve resources, each user's job is saved for a short period (30 days), allowing users to retrieve their results if the job has already been completed.

InDeep^{Holo} calculation

When an InDeep job is complete on a system, users can select specific residues to perform an InDeep^{Holo} calculation (see the InDeep^{Holo} protocol).

Server implementation

An overview of the InDeep^{Net} server implementation and the technologies employed is shown below. The content displayed on the client-side is built using the ReactJS framework (<https://react.dev/>) and incorporates the Mol* JavaScript library for protein structure visualization and interaction. The dynamic forms are formatted with the Tailwind CSS framework (<https://tailwindcss.com/>). For the backend, we developed a complete REST API using Django Rest Framework (<https://www.django-rest-framework.org/>). Input data sent to the server are securely saved in a PostgreSQL database (<https://www.postgresql.org>) using the concept of Object Relational Mapping provided by the Django framework. To perform and monitor the execution of InDeep jobs, the Django Rest Framework server communicates with the Celery (<https://docs.celeryq.dev>) task scheduler, which manages job execution. A web link is provided to the user to consult the status of their job in real-time. Finally, the results are displayed in the frontend application. If requested by the user, an email is sent when the job is completed.

A description of the server functionalities is provided in the online documentation. We also provide a step-by-step web tour tutorial to help users get started with the application. InDeep^{Net} is easy to use and completely free, eliminating the need for a complex installation of InDeep with GPU configuration.

Jobs submitted via InDeep^{Net} are executed on the Institut Pasteur Kubernetes (K8S) (<https://kubernetes.io/>) GPU cluster,

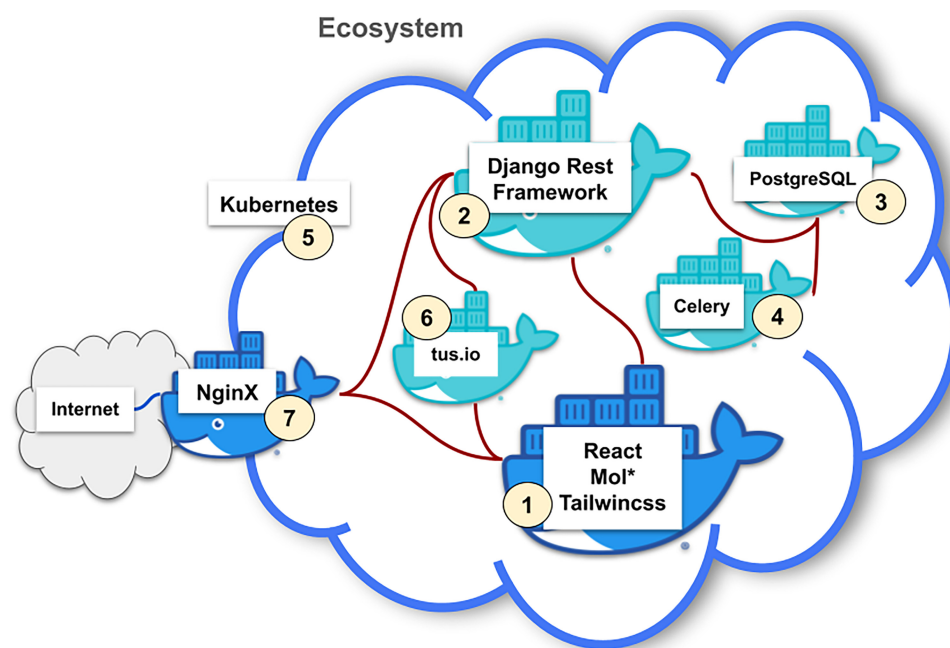


Figure 2. Ecosystem of InDeep^{Net}. The client-side uses ReactJS styled with Tailwind CSS and Mol* [1]. The backend features a REST API built with Django Rest Framework [2], utilizing PostgreSQL for data storage [3]. Job execution and monitoring are managed via Celery [4]. The server is powered by the Institut Pasteur K8S GPU cluster [5], with file transfers handled by the tus server [6]. This ecosystem is served via Nginx [7]. The code is available on GitLab.

which provides a high-performance computing environment. Typical execution times for InDeep jobs are around 15 min. The tus (<https://tus.io/>) server is used for transferring large files (typically trajectory files), and the application is served via an Nginx (<https://nginx.org>) server.

This ecosystem (Fig. 2) ensures a fast and secure user experience.

Case studies

InDeep was applied to assess ligandability and interactability across multiple protein structures, demonstrating its ability to identify drug-binding sites and PPI interfaces with high accuracy [18]. To illustrate the predictive capacity of InDeep within InDeep^{Net} Fig. 3A depicts the Ligandability channel's ability to detect structurally characterized ligand-binding sites in four distinct protein targets that fall outside the training set: PHD2 (Prolyl Hydroxylase Domain-containing Protein 2), WDR5 (WD Repeat Domain 5), MDM2 (Mouse Double Minute 2 homolog), and the SARS-CoV-2 main protease. These proteins serve as independent validation examples. In each case, InDeep^{Net} successfully identified the known binding site as the primary ligandable region, represented by red blobs, using a ligandability threshold of 0.95. These results highlight InDeep^{Net}'s ability to pinpoint potential small-molecule binding pockets, aiding drug discovery efforts targeting both enzyme active sites and protein interaction modulators. Figure 3B focuses on PPI interface identification, leveraging different InDeep^{Net} channels to analyze binding site characteristics in diverse protein complexes. The Interactability channel (purple blobs) was employed to identify key PPI regions in 7CYQ (the SARS-CoV-2 replication complex) and 2MPS (the MDM2-p53 complex), detecting the primary interaction interfaces. Additionally, the carbon alpha (CA) channel (blue blobs) and Hydrophobic channel (green blobs) were utilized to provide deeper structural insights into protein backbone localization

and hydrophobic hot spots, respectively. Specifically, in 1QHR (alpha-thrombin in complex with hirugen), InDeep^{Net} successfully identified the exosite I region, the primary binding site of hirugen. Exosite I serves as a substrate recognition site and plays a major role in thrombin's regulation of fibrinogen cleavage. Similarly, for 2MPS (MDM2-p53 complex), hydrophobic hot spots within the p53-binding cleft of MDM2 were detected, reinforcing the importance of hydrophobic interactions in stabilizing PPIs and guiding small-molecule drug design. Overall, these analyses show InDeep^{Net}'s capability to provide high-resolution structural insights into both ligandable binding sites and PPI interfaces, making it a valuable tool for structure-based drug discovery and protein interaction analysis. By enabling the identification of druggable sites, interactability regions, and hydrophobic hot spots, InDeep^{Net} supports the rational design of small-molecule inhibitors and PPI disruptors.

Conclusion and future work

InDeep^{Net} is a powerful web platform that enhances the prediction of functional binding sites in proteins, particularly for PPIs and ligandability assessments. By integrating InDeep, a deep-learning-based predictor of binding sites, with InDeep^{Holo}, a tool for evaluating holo-like conformations, InDeep^{Net} provides researchers with an advanced and accessible solution for drug discovery and structural biology. Through interactive 3D visualizations and seamless integration with Mol*, the platform facilitates in-depth analysis of binding pockets, making complex computational tools widely available without requiring specialized expertise or high-performance computing resources.

Validation across multiple datasets demonstrates that the implementation of InDeep within InDeep^{Net} represents a robust solution in accurately identifying ligandable sites and PPI

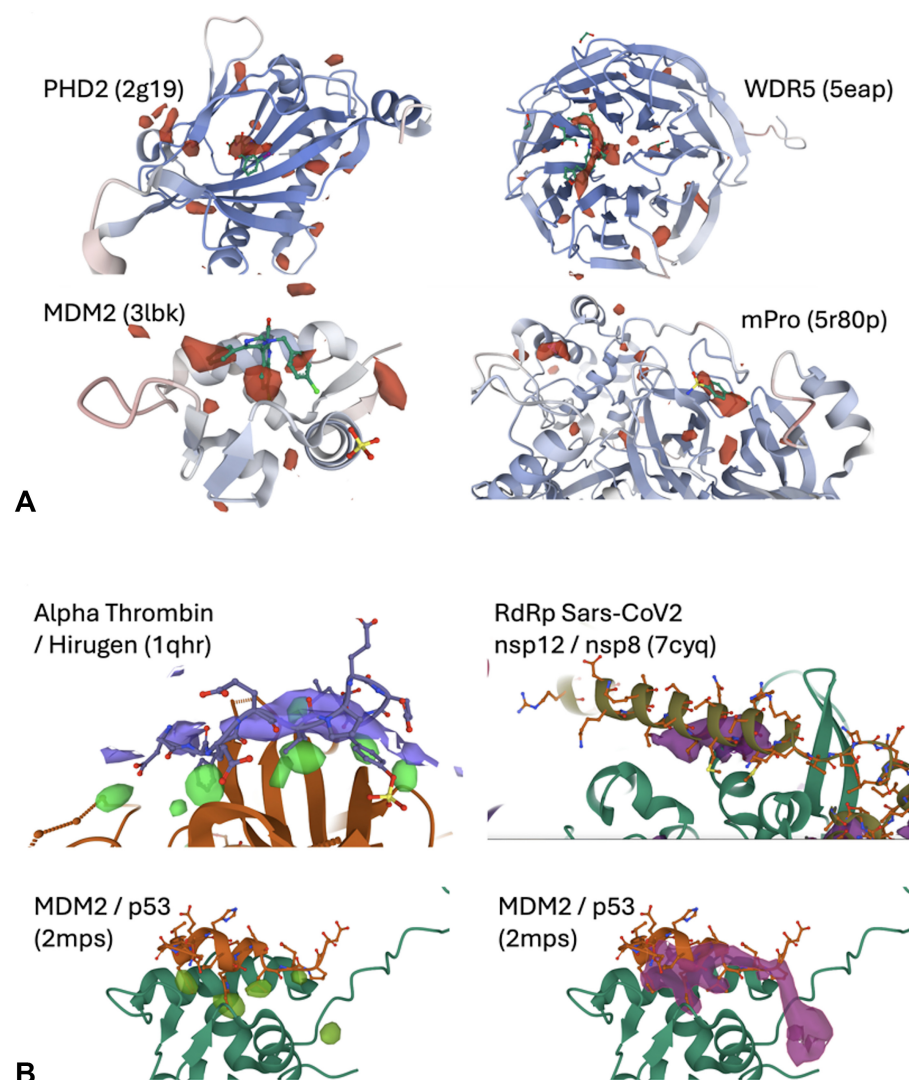


Figure 3. (A) Ligandability assessment by InDeep^{Net}. Utilization of the Ligandability channel within InDeep^{Net} to identify ligandable pockets. In all examples, namely PHD2, WDR5, MDM2, and SARS-CoV2 main protease, which are independent of the training set, InDeep^{Net} successfully identified the known binding site of structurally characterized ligands as the primary ligandable region (volume maps), using a ligandability threshold of 0.95. **(B)** Interactability assessment by InDeep^{Net}. Utilization of the Interactability channel (volume map) to identify regions mediating PPIs in structures 7CYQ and 2MPS. Additionally, the CA channel (main blobs) and Hydrophobic channel (small blobs) were used to identify backbone localization of the partner and hydrophobic hot spots in structures 1QHR and 2MPS.

interaction regions in diverse proteins. Its success in recognizing hydrophobic hotspots, protein backbone localization, and transiently ligandable pockets further reinforces its potential for structure-based drug design. By providing a publicly accessible, high-performance computing environment, InDeep^{Net} accelerates PPI-targeted therapeutic development, supporting the rational design of small-molecule inhibitors, PPI disruptors, and enzyme modulators. Future improvements will focus on allowing the use of InDeepHolo along MD trajectories to fetch more automatically pertinent pocket conformations for structure-based drug design. Moreover, we plan to make future developments of InDeep available within InDeep^{Net}.

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Supplementary data

Supplementary data is available at NAR online.

Conflict of interest

None declared.

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Data availability

InDeepNet is freely accessible at <https://indeep-net.gpu.pasteur.cloud/>.

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